BLOWER

LUBRICATION

The blower gears and bearings are lubricated by oil which drains from valve operating mechanism on cylinder head and into camshaft or balance shaft gallery pockets. After this oil reaches a certain level in pockets, it overflows through two holes at each end of blower housing, providing lubrication for blower drive gears at rear end, and governor and water-pump drives at front end. A dam in blower housing cover maintains an oil level which submerges the teeth of blower lower rotor timing gear. A slinger on opposite end of lower rotor throws oil into governor weight assembly. Surplus oil passes from blower to oil pan through drilled holes in cylinder block.

AIR BOX DRAINS

In normal operation, a slight amount of vapor from the air charge condenses and settles on the floor of the air box. This condensation is drawn off through cored passages in the ends of block and openings in the side of the engine block below the air box floor.

Pipes from the air box drain openings lead to atmosphere beneath vehicle.

Air box drains should be open at all times. Examine the openings at regular intervals and, if necessary, clean the passages. An indication of plugged drains is liquid accumulation on the air box floor. Such accumulations can be seen when removing hand hole cover plates on the side of the engine block. If accumulations appear, thoroughly wipe out air box with rags, blow out with compressed air.

CAUTION: Remove or at least loosen an air box hand hole cover, or end plate gaskets may be damaged by excessive air pressure. Remove pipe connectors from block and clean connectors and pipes, thoroughly blowing out drains with compressed air.

Under normal conditions only a small amount of waste drains from air box.

INSPECTION

Because of the important part blower plays in efficient operation of the Diesel engine, an inspection of unit at regular intervals is recommended.

When operating under adverse conditions inspection should be accomplished at more frequent intervals. If this practice is followed, minor irregularities can usually be detected and corrected before more serious difficulties develop. Every 100,000 miles, the blower should be removed, inspected, and overhauled.

A blower may be inspected for defective conditions without being removed from engine. If, however, inspection reveals that blower has been damaged or worn sufficiently to impair its efficiency, then remove from engine and either overhaul or replace.

Before examination of blower on engine for any unusual conditions, remove air cleaners, blower air inlet housing, or any other parts interfering with the air inlet system.

WARNING

1. When blower rotors are exposed and engine is in operation, keep fingers, clothing, and any loose parts away from the blower air inlet. Severe bodily injury or damage to blower may result.

2. Always disconnect starting battery cable to render engine starting equipment inoperative before starting work on a blower that involves use of the hands on blower rotors.

Scuffing Caused by Dirt or Chips

Dirt in blower invariably causes deep scratches in housing around intake and on rotor tips. Blower should be removed to bench and reconditioned. Eliminate dirt leak and service air cleaners.

Leaky Oil Seals

If oil is present on rotors or in housing, run engine at approximately 1000 rpm and observe inside of end plate with a strong light. A thin film of oil will spread around a leaking oil seal. Replace defective seals.

NOTE: Oil on rotors may also be the result of pull-over from air cleaners that are filled above oil level marks; therefore, the two conditions, oil seal leak and pull-over from air cleaners, should not be confused.

Loose Rotor Shaft or Defective Bearings

If contact has been made at any point on rotors or housing, it is an indication of rubbing and scoring between crowns or rotor lobes and mating rotor roots; between rotors and end plates; or between rotors and housing. Generally a combination of these conditions exist and are usually caused by the following:

1. A loose shaft usually causes rubbing between rotors and end plates. Worn or unsatisfactory bearings will cause rubbing between mating rotor lobes